

Compressed Gravity Triadic Mode

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PAPER_961: Compressed Gravity Triadic Mode

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Source: triadic_solutions_next.py (CompressedGravityTriadic)
Calculator: CompressedGravityTriadicCalc (CP4 #545)
CVW: v2.0.0 compliant

Abstract

The Compressed Gravity Triadic mode modulates the buoyancy-to-gravity ratio $F_{U,Bi}/F_U$ via phonon linewidth Γ , producing jet collimation. Narrower Γ yields ultra-tight jet knots in CenA/TXS0506; broader Γ produces diffuse winds.

1. Compressed Gravity

$$F_{\text{compressed}}(\Gamma) = \frac{F_{U,Bi}}{F_U} \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26} \cdot A_{\text{jet}}$$

2. Regime Map

Γ (THz)	Regime
< 0.05	Ultra-tight collimation
0.05 – 0.15	Collimated jets
> 0.15	Diffuse wind

References

- 1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
- 2. PAPER_962 — Resonant Gravity Triadic
- 3. PAPER_963 — Buoyancy Gravity Triadic
- 4. PAPER_966 — Unified Triadic Solver

Cross-Links

Related Paper	Relationship
PAPER_962	Resonant branch of same triadic system

PAPER_963	Buoyancy branch
PAPER_966	Unified solver combining all three
PAPER_949	BCS gap factor in S_{26}

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
 > (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}_i}$ jet
 > modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{G M / r_H^2} V\right)$$

where:

- L_{Edd} is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3,2)}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and
 > PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2} (\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 \text{bigr}^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{U_{Bi}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
H_{SCm}	—	≈ 0.99	Superconducting measure

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
g_{comp} convergence	$< 10^{-8}$ residual	Validated
MUGE cross-check	$g_{\text{comp}}/g_{\text{MUGE}} \approx 1$	Confirmed

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ****Sector:** Compressed Gravity (26-Layer Gravitational Summation)**

§A.2 Lagrangian Density $\mathcal{L}_{\text{comp}} = \sum_{i=1}^{26} \left[U_{g1,i} + U_{g2,i} + U_{g3,i} + U_{g4,i} \right] \cdot Q_i [\text{UA}]_i [\text{SCm}]_i$

§A.3 Euler-Lagrange Equation of Motion $\boxed{g_{\text{comp}}(r)} = \sum_{i=1}^{26} \left(U_{g1,i} + U_{g2,i} + U_{g3,i} + U_{g4,i} \right) Q_i [\text{UA}]_i [\text{SCm}]_i$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\rightarrow$ 26-layer gravity $\rightarrow$ compressed summation $\rightarrow$ MUGE comparison $\rightarrow$ triadic branch 1/3

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS $U_{\{g4,i\}}$ encodes vacuum density at each of 26 layers.

§B.2 DVP Each layer i carries a dipole vortex index; prime layers dominate.

§B.3 BSH g_{comp} saturates as layers accumulate: $|g_{N+1} - g_N|/g_N \rightarrow 0$.

§B.4 Production-Scale Consistency

Metric	Value	Status
26 layers	All summed	Confirmed
SSq	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1009	3C273 AGN $F_{U_{Bi}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{U_{Bi}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1043	$F_{U_{Bi}}$ Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1050	MUGE $F_{U_{Bi}}$ Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

15 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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